

Economy Balance Framework — Inkglass Odyssey

Design document. Read before working on relay/economy/, item schemas, recipe schemas, shop endpoints, or gathering systems.

Cross-references: docs/combat_system.md (HP tables for damage/healing scaling), docs/schemas_reference.md (item.json, recipe.json, world_config.json economy_config).

Earning Rates

Expected income per session by level band — the foundation for all pricing balance.

Level band	Session income (gc equivalent)	Source breakdown
1–5 (Newcomer)	30–80	Quest rewards 60%, gathering/vendor 30%, found coins 10%
6–10 (Established)	80–200	Quest rewards 50%, crafting sales 25%, gathering 15%, loot 10%
11–15 (Known)	200–500	Quest rewards 40%, crafting sales 30%, rare materials 20%, other 10%
16–20 (Legendary)	500–1500	Quest rewards 35%, rare crafting 35%, unique materials 20%, other 10%

Item Pricing by Type and Rarity

Type / Rarity	Common	Uncommon	Rare	Legendary
Weapons	15–50	150–500	1500–5000	Cannot purchase
Armour	20–75	200–750	2000–7500	Cannot purchase
Consumables	5–25	50–100	250–500	Cannot purchase
Tools	10–30	75–200	500–1500	Cannot purchase
Materials (per unit)	1–10	10–50	50–250	Cannot purchase

Damage Scaling by Rarity

Rarity	Typical damage dice	Average (before modifier)	Expected level band
Common	1d6, 1d8, 1d10	3.5–5.5	1–5 (Newcomer)
Uncommon	1d8+1, 1d10+1, 2d6	5.5–7	6–10 (Established)
Rare	2d6+2, 1d12+2, 2d8	9–11	11–15 (Known)
Legendary	2d8+3, 2d10+2, 3d6+2	12–15+	16–20 (Legendary)

Healing Consumable Scaling

Rarity	Healing dice	Average	Price (gc equivalent)
Common	2d4+2	7	15–25
Uncommon	4d4+4	14	50–100
Rare	8d4+8	28	250–500
Legendary	N/A — legendary consumables should have unique effects, not just bigger numbers	—	N/A

Balance principle: a common healing consumable restores roughly one hit's worth of damage from a same-tier weapon. Using a turn to heal in combat buys time but doesn't swing the fight.

Ability-based healing (spells, cultivation techniques) restores approximately 1.5–2x consumable amounts at the same tier, justified by the resource cost. Formula: healing_dice + relevant ability modifier (usually WIS/CHA equivalent).

Sell-Back Ratio

Base sell-back ratio: 50% of item value when selling to a shop NPC. Modified by faction standing: Friendly +5% (55%), Allied +10% (60%), Unfriendly -10% (40%), Hostile: shop refuses. Stored in world_config.json as sell_back_ratio (default: 0.5). The relay computes the final sell price; Unity displays it.

Transport Pricing

Standard journey fares: 5–15% of the level band’s expected session income. Premium transport (airships, aerial taxis): 2–3x standard fare. Defined per transport NPC in their transport_data field.

Equipment Repair

Repair cost: 10–20% of the item’s purchase value. Auto-calculated by the relay if not overridden in the item definition. Applied as a currency sink when equipment condition degrades (implementation deferred — Phase 3+ consideration).

Flora and Fauna Gathering Yields

An hour of gathering (3–5 successful checks) should yield 20–40% of a session’s expected income.

Gathering tier	Material value per unit	Typical yield per check	Income per hour
Common herb/ore	2–5 gc	2–4 units	12–40 gc
Uncommon herb/rare ore	10–25 gc	1–3 units	10–75 gc
Rare essence/material	50–100 gc	1 unit	50–100 gc

Fauna follows the same pattern with higher variance. Passive fauna yields common materials. Aggressive fauna (requires combat to harvest) yields uncommon or rare materials. Sapient fauna is never harvestable — they are NPCs.

Crafting Profitability

Crafting inputs must cost less than the output’s sale value. Target margin: 30–50% profit. The Workshop auto-calculates: input material cost total (from item library) versus output item value, displayed with a green/yellow/red indicator. Below 20% margin: warn. Above 60% margin: warn.

Item Types and Stats

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Item type	Stats block structure
weapon	attack_bonus (int), damage_dice (string e.g. 2d6), damage_type, range, modifier_source (STR/DEX/INT/WIS/CHA — which ability governs attack and damage), properties (array: finesse, reach, thrown, etc.)
armour	ac_bonus (int), armour_category (none/light/medium/heavy — determines DEX modifier cap), check_modifiers (map of skill: modifier)
shield	shield_bonus (int, typically +2), check_modifiers
consumable	effect_type (healing/buff/damage/utility), effect_dice (dice notation e.g. 2d4+2), duration, charges
material	crafting_categories (array), quality_tier
tool	associated_skill, proficiency_bonus_modifier
quest	associated_quest_id, description only — no mechanical stats

Item Binding

Binding type	Rule
unbound	Can be freely traded, sold, dropped
bind_on_equip	Can be traded until equipped; once equipped, locked to that character
bind_on_acquire	Locked to the acquiring character immediately (quest rewards, crafted legendaries)

Items also carry a unique boolean. If true, only one can exist per character inventory.

Gear Slots

Default slot set: head, chest, legs, hands, feet, primary_hand, secondary_hand, accessory_1, accessory_2. Worlds may override this set in world_config.json under equipment_slots. If no override is specified, the default set applies.